

Notes: Glover, Pallais & Pariente (QJE, 2017)
Discrimination as a Self-Fulfilling Prophecy: Evidence from French Grocery Stores

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Workers, stores, and schedules	4
3.2	Manager bias	4
3.3	Performance outcomes	4
3.4	Worker survey	5
4	Models and empirical specifications	5
4.1	Baseline difference-in-differences specification	5
4.2	Exogeneity of scheduled shifts	5
4.3	Mechanism tests	5
4.4	Worker-manager interaction specification	6
4.5	Minority versus majority performance and hiring logic	6
5	Identification and empirical design	6
5.1	Why the design is strong	6
5.2	Main identifying assumptions	6
5.3	Threats and robustness logic	6
6	Section-by-section study guide	7
7	Tables and figures	8
7.1	Table I: Descriptive Statistics and Table II: Exogeneity of Scheduled Shifts	8
7.2	Table III: Effect of Manager Bias on Time Spent at Work and Figure I: Manager Bias and Worker Performance	9
7.3	Table IV: Effect of Manager Bias on Performance at Work and Table V: Worker-Manager Affection and Task Assignment	10
7.4	Table VI: Worker-Manager Interaction and Table VII: Comparison of Minority and Nonminority Performance	11
7.5	Table VIII: Effect of Manager Bias on Performance in Rest of the Week	12
8	Result map	13
9	Short summary	13

10 Conclusion	14
10.1 Core conclusion	14
10.2 Economic intuition	14
10.3 Incremental contribution revisited	14
10.4 Policy implications	14
10.5 Limitations	14
10.6 Takeaway	14

1 Big picture

- **Question.** Can discriminatory beliefs held by managers directly reduce minority workers' productivity in a real workplace, creating a self-fulfilling prophecy?
- **Setting.** The paper studies entry-level cashiers in 34 outlets of a large French grocery store chain. New cashiers on six-month contracts have quasi-random schedules and work with different managers on different days.
- **Core treatment.** Managers' implicit bias is measured using an Implicit Association Test (IAT) linking North African-sounding names to worker incompetence. A worker-day has higher exposure to bias when the scheduled manager has a higher IAT score.
- **Main identification idea.** Minority and majority workers are compared within stores and across shifts, exploiting quasi-random worker-manager assignment. The key coefficient is the interaction between worker minority status and manager bias.
- **Main result.** Minority workers perform worse when scheduled with more biased managers. They are absent more often, spend less time at work, scan fewer items per minute, and take more time between customers.
- **Mechanism.** The evidence does not look like direct animus or unpleasant task assignment. It is most consistent with biased managers interacting less with minorities, and lower interaction leading to lower effort/performance.
- **Hiring implication.** Average minority and majority workers perform similarly overall, but minorities perform better than majority workers when working with unbiased managers. This is consistent with statistical discrimination in hiring: because biased managers depress minority performance, the firm may set a higher hiring threshold for minorities.
- **Economic takeaway.** Discrimination can be self-fulfilling not only through long-run human-capital investment but also through immediate workplace interactions that change worker performance.

2 Incremental contribution of the paper

- **Real workplace evidence on stereotype-induced productivity.** The paper moves beyond laboratory stereotype-threat evidence by showing that manager bias affects day-to-day performance in an actual firm.
- **A direct link from manager beliefs to worker output.** Standard statistical discrimination models often treat productivity as fixed conditional on skill. This paper shows that manager beliefs themselves can affect realized productivity.
- **Day-level within-worker variation.** The design uses worker-day observations, worker fixed effects, store fixed effects, and schedule-based assignment. This is much sharper than cross-sectional comparisons of minority and majority workers.
- **Manager-level bias measurement.** The paper links managers' implicit attitudes, measured independently using an IAT, to actual workplace outcomes.
- **Separation of mechanisms.** It distinguishes animus, stereotype threat/self-stereotyping, task assignment, worker-manager interaction, and statistical discrimination in hiring.
- **Quasi-random scheduling evidence.** Table II directly tests whether minority workers are assigned to more biased managers or worse shifts. The exogeneity evidence supports the interpretation that performance differences are caused by exposure to manager bias rather than sorting.
- **Self-fulfilling discrimination in hiring.** The paper provides a concrete microchannel through which statistical discrimination can become self-confirming: manager bias lowers minority performance, so the firm rationally infers that minority workers need higher hiring thresholds.

- **Policy relevance.** The results imply that reducing implicit bias or standardizing manager-worker interactions can affect both productivity and labor-market inequality.

3 Data and measurement

3.1 Workers, stores, and schedules

- The sample consists of entry-level cashiers hired on six-month Contrat de Professionnalisation contracts.
- These workers do not submit schedule preferences, unlike senior cashiers.
- A computer system assigns shifts to meet predicted demand while respecting senior workers' preferences.
- The final administrative sample contains 34 stores, 204 workers, and 4,371 worker-day observations.
- Workers are categorized as minority or nonminority based on names because French law limits direct ethnic data collection.

Interpretation: The schedule system is central. The empirical design is credible because new workers do not choose their managers and managers do not directly choose which new workers to supervise on a given day.

3.2 Manager bias

- Manager bias is measured using an Implicit Association Test.
- The test compares reaction times linking traditionally French or North African-sounding names to competence/incompetence words.
- The manager's IAT score is standardized.
- A worker-day exposure measure is a time-weighted average of scheduled managers' IAT scores when a worker works with multiple managers.

Interpretation: The IAT is measured outside the daily production data. This makes it useful as a predetermined manager-level bias measure rather than a variable inferred from worker outcomes.

3.3 Performance outcomes

- **Absence:** whether the worker is absent on a scheduled workday.
- **Time at work:** minutes worked in excess of scheduled time, based on time-clock data.
- **Scanning speed:** articles scanned per minute.
- **Inter-customer time:** seconds between finishing one customer's transaction and starting the next.
- **Estimated customers served:** a constructed summary measure combining time worked and transaction speed assumptions.

Interpretation: The outcomes capture both extensive-margin effort, such as coming to work and staying late, and intensive-margin effort, such as scanning and customer flow speed.

3.4 Worker survey

- After contracts ended, workers were surveyed about their interactions with managers.
- They ranked managers according to affection, promotion recommendations, enjoyment, confidence, preferred register assignments, breaks, customer-flow management, and cleaning duties.
- They also reported whether they remembered particular managers, which is used as a proxy for worker-manager interaction.

Interpretation: The survey data are essential for distinguishing mechanisms: animus, stereotype threat, task assignment, and interaction intensity.

4 Models and empirical specifications

4.1 Baseline difference-in-differences specification

The core regression is conceptually:

$$Y_{ist} = \alpha_i + \delta_s + \beta_1 \text{Minority}_i \times \text{Bias}_{st} + \beta_2 \text{Bias}_{st} + X'_{ist} \gamma + \varepsilon_{ist}.$$

- Y_{ist} is a worker-day outcome.
- Minority_i indicates minority status.
- Bias_{st} is the scheduled manager's IAT score.
- Worker fixed effects absorb time-invariant worker ability and preferences.
- Store or date/shift controls absorb local conditions.
- The coefficient of interest is β_1 .

Interpretation: β_1 measures whether minority workers are differentially affected by a more biased manager, relative to the effect of the same manager bias on nonminority workers.

4.2 Exogeneity of scheduled shifts

$$\text{ShiftCharacteristic}_{ist} = \alpha_s + \lambda \text{Minority}_i + \varepsilon_{ist},$$

with extensions including $\text{Minority}_i \times \text{Bias}_{st}$.

Interpretation: This specification checks whether minorities are systematically scheduled with different managers or different shifts. Table II supports quasi-random assignment.

4.3 Mechanism tests

$$\text{SurveyOutcome}_{im} = \alpha_i + \beta \text{Minority}_i \times \text{Bias}_m + \gamma \text{Bias}_m + u_{im}.$$

Interpretation: Survey outcomes test whether biased managers are disliked, assign unpleasant tasks, reduce confidence, or interact less with minorities.

4.4 Worker-manager interaction specification

$$Performance_{imt} = \alpha_i + \beta Remember_{im} + f(TimeShare_{im}) + \varepsilon_{imt}.$$

Interpretation: If remembering a manager proxies for more interaction, and interaction improves performance, then lower interaction by biased managers can explain the performance effects.

4.5 Minority versus majority performance and hiring logic

$$Y_{ist} = \alpha + \beta Minority_i + \delta_s + \varepsilon_{ist}.$$

Interpretation: Comparing minority and majority performance overall and on days with unbiased managers allows the paper to distinguish statistical discrimination from taste-based discrimination in hiring.

5 Identification and empirical design

5.1 Why the design is strong

- New workers have little schedule control.
- They work with different managers on different days.
- Managers' bias is measured separately through the IAT.
- Worker fixed effects compare the same worker across more and less biased managers.
- Majority workers provide a comparison group for general manager quality.

Interpretation: The main design is a within-worker, within-workplace comparison using manager bias as a rotating treatment.

5.2 Main identifying assumptions

- Minority and majority workers are not sorted to managers with different bias levels.
- Manager bias is not simply a proxy for poor general managerial ability.
- Shift characteristics do not systematically vary with minority status and manager bias.
- Performance measures capture real effort/productivity differences, not mechanical data artifacts.

Interpretation: Table II supports the first assumption. The majority-worker comparison and manager-characteristic controls support the second.

5.3 Threats and robustness logic

- Senior workers could influence schedules, but robustness checks control for store-shift timing patterns.
- More biased managers could work in busier periods, but date, shift, and store controls reduce this concern.
- Manager bias could affect senior workers rather than CPs directly, but the main patterns survive controls and heterogeneity checks.
- Survey-based mechanisms are not randomized, so they should be interpreted as supporting evidence.

Interpretation: The paper is not a randomized experiment, but the assignment environment and multiple robustness checks make the causal interpretation plausible.

6 Section-by-section study guide

- **Introduction.** Explains the theoretical distinction between taste-based discrimination, statistical discrimination, stereotype threat, and direct performance effects of discriminatory beliefs.
- **Setting.** Describes the French grocery chain, CP workers, schedule assignment, and performance measures.
- **Data and exogeneity.** Provides descriptive statistics and shows that scheduled exposure to manager bias is not different for minority workers.
- **Main results.** Shows that minority workers perform worse when scheduled with biased managers.
- **Mechanisms.** Uses survey and administrative data to test animus, task assignment, confidence, and interaction.
- **Hiring interpretation.** Shows that minorities perform better than majority workers under unbiased managers, consistent with statistical discrimination in hiring.
- **Conclusion.** Discusses self-fulfilling discrimination and possible policy responses.

7 Tables and figures

7.1 Table I: Descriptive Statistics and Table II: Exogeneity of Scheduled Shifts

Read:

- Table I reports worker, manager, and shift characteristics for the administrative and survey samples.
- Minority workers are 28 percent of the administrative sample; managers are 6 to 8 percent minority depending on sample.
- About 66 percent of managers have moderate-to-severe IAT bias and 9 percent little-to-no bias.
- Table II shows that minority workers are not systematically assigned to more biased managers or worse shifts.
- The minority worker by manager bias interactions in Table II are small and insignificant across schedule characteristics.

Detailed interpretation: Table I establishes the empirical population and highlights the extent of implicit bias among managers. Table II is a critical identifying table. If minorities were systematically assigned to more biased managers, the main estimates could reflect sorting. Instead, the table shows that scheduled exposure to manager bias and other scheduled shift features are balanced by minority status.

Economic message: The first table defines the treatment environment, while the second supports the quasi-random assignment needed for causal interpretation.

TABLE I
DESCRIPTIVE STATISTICS

	Administrative data sample		Worker survey sample	
	All observations from included stores	Regression sample (observations with manager IAT scores)	All survey respondents	Survey respondent regression sample
Panel A: Worker characteristics				
Minority	28%	28%	29%	25%
Male	6.9%	7.4%	7.7%	7.3%
Age			29.9	30.1
Number of previous jobs			3.9	4.0
Less than high school education			58%	61%
High school degree			35%	32%
More than high school education			7%	7%
Number of workers	218	204	310	178
Panel B: Manager characteristics				
Minority	6%	8%		
Male	10%	7%		
Level 4 (high position)	18%	18%		
Age	41.1	41.1		
Average IAT score (in standard devs)		1.35		
Moderate to severe bias		66%		
Slight bias		20%		
Little to no bias		9%		
Preference for minorities		4%		
Number of managers	154	119		
Panel C: Shift characteristics				
Scheduled days per week	4.2	4.2		
Scheduled hours per day	7.2	7.2		
Absent	1.8%	1.6%		
Minutes worked in excess of schedule	-0.31	-0.06		
Articles scanned per minute	18.5	18.5		
Inter-customer time (seconds)	29.2	28.7		
Payment time (seconds)	50.7	50.8		
Number of shifts	5,099	4,371		
Number of stores	34	34	70	51

Note. The first two columns of data provide descriptive statistics for the sample for whom we have administrative data. The first includes all observations from the 34 included stores, while the second includes only observations for which we have the manager's IAT score. The final two columns provide descriptive statistics for the worker survey sample. The first includes all survey respondents, while the second includes only those workers for whom we either have managers' IAT scores or performance data and are thus included in the analysis. Level 4 managers have a higher position in the store than the remaining managers. Manager age is reported as of January 1, 2012. *Moderate to severe bias* is defined as having a raw IAT score above 0.35, *Slight bias* is defined as having a raw IAT score between 0.15 and 0.35, *Little to no bias* is defined as having a raw IAT score between -0.15 and 0.15, and *Preference for minorities* is defined as having a raw IAT score below -0.15.

(a) Table I: Descriptive Statistics.

TABLE II
EXOGENEITY OF SCHEDULED SHIFTS

	Manager bias	Minority manager	Male manager	Level 4 manager	Total managers	Articles per minute in other stores on date	Shift includes early morning	Shift includes late evening	Total hours
Panel A: Minority workers									
Minority worker	0.005 (0.022)	0.000 (0.063)	-0.011 (0.009)	-0.001 (0.004)	-0.025 (0.043)	0.045 (0.086)	0.014 (0.015)	0.021 (0.013)	0.017 (0.034)
Panel B: Minority workers and manager bias									
Minority worker × manager bias	0.009 (0.007)	-0.002 (0.008)	0.008 (0.010)	-0.013 (0.024)	0.049 (0.044)	-0.004 (0.008)	0.007 (0.007)	0.038 (0.037)	
Minority worker	-0.009 (0.009)	-0.009 (0.007)	-0.010 (0.012)	-0.011 (0.034)	-0.007 (0.074)	0.019 (0.017)	0.013 (0.013)	-0.024 (0.049)	
Manager bias	-0.032 (0.021)	0.015 (0.025)	0.049 (0.033)	0.066 (0.048)	-0.043 (0.093)	-0.002 (0.011)	-0.003 (0.019)	-0.058 (0.062)	
Observations	4,371	4,371	4,371	4,371	4,238	4,371	4,371	4,368	
Dependent variable mean	1.13	0.060	0.114	0.171	2.74	18.23	0.141	0.580	7.22
Store fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note. Each column in Panel A shows the result of regressing the dependent variable indicated by the column heading on an indicator for the worker being a minority. Each in Panel B shows the results of regressing the same dependent variable on a dummy for the worker's minority status, the manager's IAT score (in standard deviation terms), the interaction of the worker's minority status and the manager's IAT score. Both the dependent variables and the manager's IAT score are based on the store's schedule, no realizations. For example, *Shift includes early morning* is a dummy variable for the shift being scheduled to start at 8 am or earlier, regardless of whether the worker arrived time. *Shift includes late evening* is an indicator for the shift being scheduled to end at 8 pm or later. *Manager bias* is the manager's IAT score, while *Minority manager*, *Male n* and *Level 4 manager* are indicators for a manager being a minority, being male, and having a high-level management position, respectively. When workers are scheduled with more than one manager, manager variables are averages, weighted by the amount of time workers were scheduled to work with each manager. Observations are weighted by the amount of time workers were scheduled to work with each manager.

(b) Table II: Exogeneity of Scheduled Shifts.

7.2 Table III: Effect of Manager Bias on Time Spent at Work and Figure I: Manager Bias and Worker Performance

Read:

- Table III shows that minority workers are more likely to be absent when scheduled with more biased managers.
- The minority-worker by manager-bias coefficient is about 0.01 in the absence regressions, large relative to the mean absence rate of 1.6 percent.
- Minority workers also work fewer minutes beyond schedule when exposed to biased managers.
- Figure I plots the raw relationship between manager IAT scores and scanning speed for minority and majority workers.
- The figure shows a negative relationship for minorities and a flatter or slightly positive relationship for majority workers.

Detailed interpretation: Table III is the first main causal result. It shows that manager bias affects whether minorities show up and how long they stay. The figure is not a causal specification, but it visually summarizes the same qualitative pattern for scanning speed. Together, they show that bias affects both time at work and on-the-job performance.

Economic message: Biased managers reduce minority labor supply and effort in measurable ways, reducing paid time and productivity.

TABLE III
EFFECT OF MANAGER BIAS ON TIME SPENT AT WORK

Panel A: Dependent variable: absence indicator				
Minority worker × manager bias	0.0098** (0.0039)	0.0095** (0.0040)	0.0117*** (0.0042)	0.0118*** (0.0043)
Manager bias	−0.0021 (0.0031)	−0.0021 (0.0032)	−0.0050 (0.0040)	−0.0052 (0.0042)
Minority worker × minority manager				0.0081 (0.0972)
Minority manager				−0.0057 (0.0153)
Observations	4,371	4,371	4,371	4,371
Dependent variable mean	0.0162	0.0162	0.0162	0.0162
R-squared	0.0005	0.0031	0.0835	0.0835
Panel B: Dependent variable: minutes worked in excess of schedule				
Minority worker × manager bias	−3.295** (1.550)	−3.279** (1.588)	−3.327* (1.687)	−3.237* (1.678)
Manager bias	−0.002 (1.141)	−0.002 (1.167)	−0.005 (0.969)	−0.005 (1.009)
Minority worker × minority manager				0.349 (10.501)
Minority manager				−3.712 (4.592)
Observations	4,163	4,163	4,163	4,163
Dependent variable mean	−0.068	−0.068	−0.068	−0.068
R-squared	0.001	0.008	0.129	0.129
Individual fixed effects	Yes	Yes	Yes	Yes
Day of the week fixed effects	No	Yes	No	No
Morning/evening fixed effects	No	Yes	Yes	Yes
Date fixed effects	No	No	Yes	Yes

Note. Each column in each panel shows the result of regressing the dependent variable on the interaction of the worker's minority status and the manager's IAT score (in standard deviation terms), controlling for the manager's IAT score and worker fixed effects. The dependent variable in Panel A is an indicator for the worker being absent. The dependent variable in Panel B is the number of minutes worked in excess of the number of minutes the worker was scheduled to work. The first column includes no additional controls. The second

(a) Table III: Time Spent at Work.

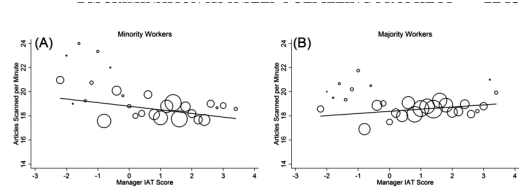


FIGURE I
Manager Bias and Worker Performance
The size of each marker indicates the number of observations in the bin.

(b) Figure I: Manager Bias and Worker Performance.

7.3 Table IV: Effect of Manager Bias on Performance at Work and Table V: Worker-Manager Affection and Task Assignment

Read:

- Table IV shows that minority workers scan fewer articles per minute and take longer between customers when assigned to more biased managers.
- The scanning-speed effect is around -0.25 to -0.28 articles per minute per standard deviation of manager bias.
- Table V finds little evidence that biased managers are disliked by minority workers or that they trigger conscious confidence loss.
- Table V, Panel B shows biased managers are less likely to assign minority workers to cleaning duties, inconsistent with a simple unpleasant-task animus story.

Detailed interpretation: Table IV is the main intensive-margin productivity evidence. Table V then evaluates mechanisms. If animus or task punishment drove the result, minority workers should report worse affection outcomes or worse task assignments under biased managers. The evidence does not support that interpretation.

Economic message: Manager bias lowers minority productivity, but not because managers obviously punish minority workers with unpleasant tasks.

TABLE IV
EFFECT OF MANAGER BIAS ON PERFORMANCE AT WORK

Panel A: Dependent variable: articles scanned per minute				
Minority worker × manager bias	−0.276** (0.109)	−0.279** (0.111)	−0.233** (0.108)	−0.249** (0.111)
Manager bias	0.140* (0.083)	0.140 (0.083)	0.080 (0.065)	0.102 (0.073)
Observations	3,601	3,601	3,601	3,601
Dependent variable mean	18.53	18.53	18.53	18.53
R-squared	0.001	0.013	0.195	0.195
Panel B: Dependent variable: inter-customer time (seconds)				
Minority worker × manager bias	1.213** (0.590)	1.228** (0.553)	1.417** (0.649)	1.360** (0.665)
Manager bias	−0.648 (0.386)	−0.571 (0.376)	−0.656 (0.521)	−0.580 (0.534)
Observations	3,287	3,287	3,287	3,287
Dependent variable mean	28.70	28.70	28.70	28.70
R-squared	0.001	0.013	0.195	0.195
Individual fixed effects	Yes	Yes	Yes	Yes
Day of the week fixed effects	No	Yes	No	No
Morning/evening fixed effects	No	Yes	Yes	Yes
Date fixed effects	No	No	Yes	Yes
Manager minority variables	No	No	No	Yes

Note. Each regression shows the result of regressing the dependent variable on the interaction of the worker's minority status and the manager's IAT score (in standard deviation terms), controlling for the manager's IAT score and worker fixed effects. The dependent variables are the number of articles scanned per minute (Panel A) and the average number of seconds between finishing one customer's transaction and starting to scan the next customer's items (Panel B). The first column includes no controls. The second column adds day of the week fixed effects, an indicator for the shift starting at 9 am or earlier, and an indicator for the shift ending at 8 pm or later. The third column includes date fixed effects and drops the day of the week fixed effects. The last column adds a dummy for the manager being a minority and the interaction of the worker's and the manager's minority status. Observations are worker-days and standard errors are clustered at the store level. *, ** denote significance at the 10% and 5% levels, respectively.

(a) Table IV: Performance at Work.

TABLE V
WORKER-MANAGER AFFECTION AND TASK ASSIGNMENT

Panel A: Worker-manager affection				
	Manager liked you best	Manager most likely to recommend you for promotion	You enjoyed working with manager best	Manager initially made you feel most confident
Minority worker × manager bias	0.019 (0.246)	0.078 (0.212)	0.243 (0.234)	0.194 (0.196)
Manager bias	0.152 (0.131)	0.251* (0.148)	−0.061 (0.162)	0.134 (0.127)
Observations	3,036	2,862	3,209	3,189
Dependent variable mean	3.991	4.053	4.062	4.073
R-squared	0.015	0.042	0.010	0.026
Panel B: Task assignment				
	Manager assigned to preferred register type	Manager assigned best breaks	Management of lines and customer flows encouraged performance	Manager assigned to fewest cleaning duties
Minority worker × manager bias	−0.035 (0.391)	0.146 (0.469)	−0.153 (0.308)	0.673*** (0.189)
Manager bias	0.021 (0.157)	−0.083 (0.146)	0.129 (0.137)	−0.276 (0.182)
Observations	2,288	2,553	2,864	2,235
Dependent variable mean	4.010	3.922	4.215	3.373
R-squared	0.002	0.008	0.018	0.045

Note. Each column in each panel shows the result of regressing the dependent variable on the interaction of the worker's minority status and the manager's IAT score (in standard deviation terms), controlling for the manager's IAT score and worker fixed effects. The dependent variable is the worker's ranking of the manager on the question indicated by the column heading. This ranking ranges from 1 (the lowest ranked manager) to N (the highest ranked manager), where N is the number of managers the worker had. Observations are

(b) Table V: Affection and Task Assignment.

7.4 Table VI: Worker-Manager Interaction and Table VII: Comparison of Minority and Nonminority Performance

Read:

- Table VI shows that workers scan more articles per minute when they remember the manager, consistent with interaction improving performance.
- Minority workers are less likely to remember more biased managers, suggesting lower interaction.
- Table VII shows that average minority and nonminority performance is similar overall.
- On days with unbiased managers, minority workers perform better: lower absence, faster scanning, lower inter-customer time, and more estimated customers served.

Detailed interpretation: Table VI provides the preferred mechanism: less interaction. Biased managers appear not to treat minorities overtly badly but interact with them less, reducing effort. Table VII links this performance effect to hiring: if minority workers perform better under unbiased managers but similarly on average, the firm may have selected minorities with higher underlying productivity to offset expected bias exposure.

Economic message: The workplace effect of bias can make statistical discrimination self-confirming: bias reduces minority output, and the firm responds by holding minorities to a higher hiring bar.

TABLE VI
WORKER-MANAGER INTERACTION

Panel A: Effect of worker-manager interaction on performance				
	Dependent variable: articles scanned per minute			
Remembers manager (indicator)	1.510** (0.635)		1.587** (0.630)	2.053*** (0.744)
Fraction of time scheduled with manager		-1.172 (1.352)	1.724 (3.638)	4.021 (3.828)
(Fraction of time scheduled with manager) ²			-4.454 (4.886)	-6.603 (5.407)
Individual fixed effects	Yes	Yes	Yes	Yes
Manager fixed effects	No	No	No	Yes
Observations	1,885	1,885	1,885	1,885
Dependent variable mean	18.42	18.42	18.42	18.42
R-squared	0.008	0.001	0.010	0.095
Panel B: Minority status, manager bias, and worker-manager interaction				
	Dependent variable: remembers manager (indicator)	Dependent variable: articles scanned per minute		
	All days	All days	Days where worker remembers manager	
Minority worker × manager bias	-0.0152* (0.0086)	-0.415* (0.209)	-0.311 (0.314)	
Manager bias	0.0190* (0.0097)	0.271** (0.114)	0.203** (0.095)	
Fraction of time scheduled with manager	0.6362* (0.3351)	-1.932 (4.159)	-5.115 (4.161)	
(Fraction of time scheduled with manager) ²	-0.5605 (0.3981)	1.017 (6.270)	4.114 (5.757)	
Individual fixed effects	Yes	Yes	Yes	
Observations	3,958	1,584	1,317	
Dependent variable mean	0.932	18.52	18.66	
R-squared	0.017	0.006	0.005	

Note. Each regression in Panel A shows the results of regressing *articles scanned per minute* on the variables listed in the left-most column, controlling for worker fixed effects. *Remembers manager* is an indicator for the worker reporting in the worker survey that she remembered the manager she was scheduled to work with that day.

(a) Table VI: Worker-Manager Interaction.

TABLE VII
COMPARISON OF MINORITY AND NONMINORITY PERFORMANCE

	Absence indicator	Minutes worked in excess of schedule	Articles scanned per minute	Inter-customer time (seconds)	Estimated customers served
Panel A: All days					
Minority worker	-0.0041 (0.0072)	0.522 (2.213)	0.282 (0.329)	0.504 (0.719)	2.80 (2.02)
Nonminority mean	0.0187	-1.186	18.55	28.21	162
Observations	4,371	4,163	3,601	3,287	3,086
Store fixed effects	Yes	Yes	Yes	Yes	Yes
Panel B: Days with unbiased managers					
Minority worker	-0.0127* (0.0067)	2.572 (2.331)	0.745** (0.323)	-2.075* (1.113)	13.94** (4.84)
Nonminority mean	0.0267	-4.268	18.65	26.59	162
Observations	482	444	367	330	301
Store fixed effects	Yes	Yes	Yes	Yes	Yes
Panel C: Days with biased managers					
Minority worker	-0.0047 (0.0094)	0.271 (2.872)	0.006 (0.383)	0.936 (0.935)	2.21 (2.68)
Nonminority mean	0.0194	-1.106	18.65	27.94	162
Observations	3,474	3,319	2,832	2,555	2,395
Store fixed effects	Yes	Yes	Yes	Yes	Yes
Panel D: All other days					
Minority worker	0.0026 (0.0027)	0.379 (1.625)	1.292** (0.325)	0.178 (0.186)	-0.25 (2.05)
Nonminority mean	0.0036	0.928	17.69	31.88	156
Observations	445	429	422	421	410
Store fixed effects	Yes	Yes	Yes	Yes	Yes

Note. Each column in each panel shows the result of a separate regression of the dependent variable indicated by the column on an indicator for the worker being a minority, controlling for store fixed effects. *Estimated customers served* is calculated under the assumptions that customers average 25 items. Standard errors are clustered at the store level. *Days with unbiased managers* are days where the worker spent at least 15 minutes with the manager.

(b) Table VII: Minority and Nonminority Performance.

7.5 Table VIII: Effect of Manager Bias on Performance in Rest of the Week

Read:

- Table VIII tests whether workers compensate later in the week after performing poorly with biased managers.
- If minorities were simply shifting effort across days, working with a biased manager should predict better rest-of-week performance.
- The table finds no evidence of compensating effort.
- For scanning speed, the coefficient is negative, suggesting possible persistence or cumulative effects rather than intertemporal substitution.

Detailed interpretation: Table VIII addresses a key alternative explanation. The main results could be less concerning if minorities merely conserved effort on biased-manager days and made it up later. The table rejects that interpretation. Bias exposure appears to lower total performance rather than simply reallocating effort over time.

Economic message: Manager bias produces real productivity loss, not just timing substitution in worker effort.

TABLE VIII
EFFECT OF MANAGER BIAS ON PERFORMANCE IN REST OF THE WEEK

	Dependent variable: average performance in rest of the week			
	Absence indicator	Minutes worked in excess of schedule	Articles scanned per minute	Inter-customer time (seconds)
Minority worker $\times$ manager bias	0.0023 (0.0020)	-0.628 (0.940)	-0.099** (0.041)	-0.336 (0.274)
Manager bias	-0.0029 (0.0017)	1.185* (0.638)	0.063 (0.046)	-0.030 (0.203)
Observations	4,271	4,174	3,935	3,610
Dependent variable mean	0.0153	-0.053	18.43	29.04
R-squared	0.0010	0.0058	0.0061	0.0026

Notes. Each column shows the result of regressing the workers' average performance in the rest of the week on the interaction of the worker's minority status and the manager's IAT score (in standard deviation terms) on a given day. Regressions control for the manager's IAT score on that day and worker fixed effects. They also control for the average manager IAT score in the rest of the week and this score interacted with the worker's minority status. Standard errors are clustered at the store level. *, ** denote significance at the 10% and 5% levels, respectively.

Table VIII: Effect of Manager Bias on Performance in Rest of the Week.

8 Result map

- **Balance/exogeneity:** minority workers are not scheduled with different manager types or shift conditions in a way that explains the results.
- **Extensive margin:** biased managers increase minority absences and reduce time at work.
- **Intensive margin:** biased managers reduce minority scanning speed and increase time between customers.
- **Animus/task-assignment mechanism:** little support; minorities do not report worse affection and are not assigned more unpleasant tasks.
- **Interaction mechanism:** strong support; biased managers interact less with minorities and interaction predicts better performance.
- **Hiring implication:** minorities perform better under unbiased managers but similarly on average, consistent with statistical discrimination in hiring standards.
- **Effort substitution test:** no evidence that minorities make up lost effort later in the week.

9 Short summary

- The paper studies new cashiers in 34 French grocery stores.
- Manager bias is measured using an IAT linking North African names to worker incompetence.
- New workers have quasi-random schedules and work with different managers across days.
- Minority workers perform worse when assigned to more biased managers.
- Effects appear on absence, time at work, scanning speed, and inter-customer time.
- The mechanism is not overt animus or unpleasant task assignment.
- Biased managers appear to interact less with minority workers, and interaction improves performance.
- Minority workers perform better than majority workers on days with unbiased managers.
- This is consistent with statistical discrimination in hiring induced by manager bias.
- The paper shows discrimination can be self-fulfilling through immediate workplace interactions.

10 Conclusion

10.1 Core conclusion

Glover, Pallais, and Pariente show that manager bias has real productivity consequences. Minority cashiers perform worse when scheduled with biased managers, even though scheduled exposure to those managers is quasi-random. Bias affects both attendance/time at work and on-the-job speed.

10.2 Economic intuition

Manager bias can reduce performance without explicit punishment. A biased manager may interact less, monitor less, ask for fewer extra tasks, or provide less encouragement. Workers notice lower interaction and respond with lower effort. The resulting lower productivity then confirms the firm’s belief that minority workers are less productive.

10.3 Incremental contribution revisited

The paper’s major contribution is showing a direct causal channel from discriminatory beliefs to worker performance inside a firm. It extends statistical discrimination theory by showing that employer beliefs can affect realized productivity, not just beliefs about productivity or long-run skill investment.

10.4 Policy implications

Bias-reduction policies may improve productivity, not only equity. But the paper also suggests that reducing the behavioral consequences of bias may require managerial routines: equal interaction, standardized monitoring, transparent task assignment, and structured feedback.

10.5 Limitations

The study concerns entry-level cashiers in one French grocery chain. The IAT captures implicit association with North African names and competence, not all dimensions of discrimination. Survey-based mechanism evidence is supporting evidence rather than randomized treatment variation. The paper also lacks direct data on contract renewal decisions.

10.6 Takeaway

Discrimination can be self-fulfilling in the short run. Biased managers can lower minority worker performance through everyday interactions, causing minorities to appear less productive and sustaining discriminatory hiring beliefs.